

Project Design Phase

Problem – Solution Fit Template

Date	13 July 2026
Team ID	SWUID2026196344
Project Name	Nutrition Assistant
Maximum Marks	2 Marks

Problem – Solution Fit Template:

Problem–Solution Fit is the process of validating that the proposed solution effectively addresses the real problems experienced by the target users. It ensures that the application is designed based on actual user needs rather than assumptions, resulting in a practical, user-friendly, and valuable solution.

For the Nutrition Assistant project, the Problem–Solution Fit focuses on helping users overcome the challenges of meal planning, nutrition tracking, and maintaining healthy eating habits. The application provides personalized meal plans, nutrition suggestions, daily nutrition monitoring, and secure user management through a responsive MERN Stack web application.

Purpose:

- ☑ Understand the nutritional challenges faced by users and provide practical solutions through a personalized web application.
- ☑ Simplify meal planning and daily nutrition tracking using an intuitive and easy-to-use interface.
- ☑ Provide personalized nutrition suggestions that help users make healthier food choices based on their preferences and goals.
- ☑ Enable users to securely manage their profiles, meal plans, and nutrition records using JWT-based authentication and centralized data storage.
- ☑ Promote healthier lifestyles by encouraging consistent nutrition tracking and informed dietary decisions through a responsive MERN Stack application.

Template:

1. CUSTOMER SEGMENT(S)CS <i>Who is your customer?</i> - Health-conscious individuals - Students and working professionals - Fitness enthusiasts - People looking to improve eating habits - Individuals with specific diet goals (weight loss, gain, maintenance)	6. CUSTOMER CONSTRAINTSCC <i>What constraints prevent your customers from taking action or limit their adoption of solutions?</i> - Busy lifestyle and lack of time for meal planning - Difficulty in calculating nutrition and calories - Lack of personalized meal recommendations - Confusion about what to eat for health goals - Forgetting to track daily nutrition - Limited access to dietitian or expert guidance	5. AVAILABLE SOLUTIONSAS <i>What solutions are available for your customers that they face today?</i> - Generic diet charts and plans from websites - Health and fitness mobile apps - Manual tracking using notebooks / spreadsheets - Consultation with dietitians (offline/online) - YouTube and social media nutrition advice - Food delivery services with healthy options
2. JOBS-TO-BE-DONE / PROBLEMSJ&P <i>Which jobs-to-be-done (or problems) do you address for your customers?</i> - Need help in planning balanced meals daily - Don't know nutritional value of food - Hard to track calories, protein, carbs, and fat - Lack of personalized meal suggestions - Difficulty in maintaining healthy eating habits - Need motivation and reminders for consistency	9. PROBLEM ROOT CAUSERC <i>What is the real reason that this problem exists? What is the root cause behind the need to do this job?</i> - Lack of awareness about nutrition - No easy way to plan and track nutrition - Information is scattered and not personalized - Time constraints and busy schedules - No centralized platform for tracking and guidance - Inconsistent habits due to lack of reminders	7. BEHAVIOURBE <i>What does your customer do to address this problem and get the job done?</i> - Search for diet plans and meal ideas online - Skip meals or eat whatever is convenient - Track food occasionally (if at all) - Follow advice from friends, family or influencers - Use multiple apps or tools separately - Start with motivation but not consistent
3. TRIGGERSTR <i>What triggers customers to act? i.e. seeing their neighbour meeting their goals, reading about a new diet trend, doctor's advice, etc.</i> - Seeing others' fitness transformations - Doctor's advice or health reports - Starting a fitness or weight loss goal - Social media and fitness content - Feeling tired, unhealthy or low energy	10. YOUR SOLUTIONSL <i>If you are working on an existing business, write about your current solution here.</i> - Nutrition Assistant is a MERN Stack web application that helps users plan meals, get nutrition suggestions, and track daily nutrition. - Personalized meal plans based on goals and preferences - Nutrition suggestions with calories, protein, carbs, fat - Track daily meals and nutrition intake - Dashboard to view insights and progress - Secure authentication and profile management - Reminders and recommendations for healthy habits	8. CHANNELS OF BEHAVIOURCH <i>8.1 ONLINE</i> <i>What kind of actions do customers take online? (search online channels from?)</i> - Google search, health blogs, YouTube - Social media platforms (Instagram, YouTube, etc.) - Health and fitness apps - Online communities and forums <i>8.2 OFFLINE</i> <i>What kind of actions do customers take offline? (search offline channels from?)</i> - Consult dietitians or doctors - Follow advice from family and friends - Read books, magazines, or brochures - Attend fitness centers or workshops